Longevity-associated orthologs across major model organisms

Six selected families • Primary lifespan experiments • Explicit OMA relationships

Arrows show the reported effect of the stated intervention; they are not GenAge Pro/Anti labels.

	Budding yeast	Nematode	Fruit fly	Mouse
TOR 6 direct OMA pairs	**TOR1** Deletion **Replicative life ↑** PMID 16293764	**let-363** RNAi **Adult life ↑** PMID 14668850	**Tor** Dominant-negative **Adult life ↑** PMID 15186745	**Mtor** Hypomorph **Median survival ↑** PMID 23994476
Direct edges: All six species pairs; yeast relationships are one-to-many.				
S6 kinase 2 direct OMA pairs	Not selected	**rsks-1** RNAi **Mean life ↑** PMID 17266680	**S6k** Dominant-negative **Adult life ↑** PMID 15186745	**Rps6kb1** Global knockout **Female life ↑*** PMID 19797661
Direct edges: Worm–fly and fly–mouse; no direct worm–mouse pair in the frozen selection.				
Insulin/IGF receptor 2 direct OMA pairs	Not selected	**daf-2** Hypomorph **Adult life ↑** PMID 8247153	**InR** Hypomorph **Female life ↑** PMID 11292875	**Igf1r** Heterozygous KO **Female life ↑*†** PMID 12483226
Direct edges: Worm–fly and worm–mouse; Igf1r is one of two mouse receptor paralogs.				
AMPK 2 direct OMA pairs	**SNF1** Overexpression **Replicative life ↓** PMID 10921902	**aak-2** Extra gene copies **Mean life ↑** PMID 15574588	**SNF1A** Adult tissue expression **Adult life ↑** PMID 23312286	Not selected
Direct edges: Yeast–fly and worm–fly; no direct yeast–worm pair in the frozen selection.				
ATG7 3 direct OMA pairs	**ATG7** Deletion **Chronological life ↓** PMID 19302372	**atg-7** RNAi in eat-2 **DR benefit reduced‡** PMID 17912023	**Atg7** Null mutation **Adult life ↓** PMID 18056421	Not selected
Direct edges: All three pairs among yeast, worm and fly; each is one-to-one.				
SOD2 6 direct OMA pairs	**SOD2** Deletion in sch9Δ **CLS benefit lost§** PMID 12586694	**sod-2** Deletion **Adult life ↑** PMID 19197346	**Sod2** RNAi **Adult life ↓** PMID 12456885	**Sod2** Connective-tissue KO **Survival ↓** PMID 21108731
Direct edges: All six species pairs; worm relationships are one-to-many.				

* Mouse S6K1/Igf1r benefits shown are female-specific; corresponding male increases were not significant.

† The original mouse Igf1r increase was not reproduced in a later cohort (PMID 22132081).

‡ Worm ATG7: reduced longevity benefit in dietary-restricted eat-2 mutants; timing and background matter.

§ Yeast SOD2: loss of sch9Δ chronological lifespan extension. Mouse SOD2: connective-tissue deficiency.

AMPK and SOD2 illustrate context-dependent results. Yeast replicative and chronological life differ.

TOR/S6K share a pathway. Rows are selected examples, not independent mechanisms or OMA orthogroups.

Empty cells mean not selected, not absence of an ortholog or effect. See the legend for missing direct edges.

Sources: GenAge Build 21, OMA All.May2026 and cited primary studies; full catalogue: 206 overlapping pairs.